Attendance Code: 564282



UNIVERSITY OF LINCOLN

CMP2028M

Natural Language Processing

Week 6: “Question Answering & Information Retrieval”

Heriberto Cuayahuitl

https://staff.Lincoln.ac.uk/hcuayahuitl

Agenda

- Week 1, 5-6 Feb: Tokenization and regular expressions
- Week 2, 12-13 Feb: N-gram language models
- Week 3, 19-20 Feb: Text classification
- Week 4, 27-27 Feb: Vector semantics and embeddings
- Week 5, 5-6 Mar: Large Language Models (LLMs)
- Week 6, 12-13 Mar: **Question answering & information retrieval**
- Week 7, 19-20 Mar: Constituent and dependency parsing
- Week 8, 26-27 Mar: Information extraction
- Week 9, 23-24 Apr: Overview of neural networks
- Week 10, 30 Apr-1May: Machine translation
- Week 11, 7-8 May: Chatbots and dialogue systems
- Week 12, 14-15 May: Speech recognition and synthesis

Easter break

Last Week

Overview of the AI behind LLMs

Sampling for language generation

- Top-k, top-p, temperature sampling

Prompting

- Zero/few-shot, CoT, In-Context Learning (ICL)

Prompt chaining is another type of prompting.

Issues with LLMs

- Hallucinations, toxic language, malicious use

Today

- **Question Answering**
- Information Retrieval
- Retrieval Augmented Generation

Acknowledgement: Some images from https://web.stanford.edu/~jurafsky/slp3/11.pdf

Types of Question Answering

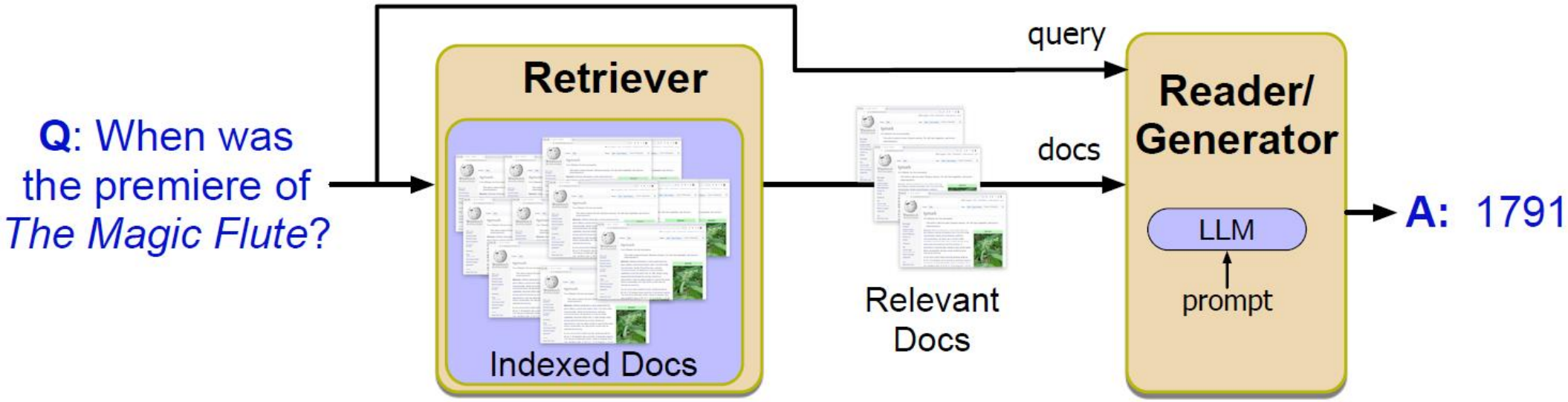
1. Response type:
   - Factoid/Binary/Numeric/Definition QA
   - Multi-choice QA
   - Non-Factoid QA
   - Multi-hop QA
2. Input type:
   - Open/Closed Domain QA
   - Knowledge-based QA
   - Text/Spoken/Visual/Multimodal QA
3. System architecture:
   - Extractive/Abstractive QA
   - Retrieval-Augmented QA
   - Conversational QA

But questions and answers may combine those three types. Here a possible/sensible combination:

Binary + Open-Domain + Knowledge-Based + Text-Based QA + **Retrieval-Augmented + Abstractive architecture**.

Example: “Is Mount Everest the tallest mountain in the world?”

Question Answering (QA) via Retrieval Augmented Generation



Retriever: returns relevant documents from a collection.

Generator: provides a response given a prompt and relevant docs.

Note that using RAG corresponds to an OPEN book QA task and not using RAG to a CLOSED book QA task.

Question Answering Datasets

- **Natural Questions**: a set of 300K anonymised English question-answer pairs to the Google search engine.
- **MS Marco**: a collection of datasets, including 1M real anonymised English questions from MS Bing queries.
- **TyDi QA**: 204K question-answer pairs from 11 languages, including Arabic, Bengali, Kiswahili, Russian, and Thai.
- **MMLU** (Massive Multitask Language Understanding): commonly-used dataset of 15908 knowledge and reasoning questions in 57 areas including medicine, mathematics, computer science, law, and others.

Visual Question Answering (VQA)



Q : Are there shadows on the sidewalk?
A : Yes

(a) Yes/No



Q : What kind of bears are these? A : 1
1. Brown 2. Grizzly 3. Polar

(b) Multi-choice



Q : How many windows can you see?
A : 1

(c) Counting



Q : What is the man doing in the street?
A : Walking

(d) Open-ended

Kim, et al. "Visual-Question Answering: A Survey of Methods, Datasets, Evaluation, and Challenges". ACM Computing Surveys, 2025.

Evaluation of QA Systems

- **Exact Match (EM)**: The percentage of predicted answers that match the gold answer exactly.

or

if BERTScore(pred, ref) > 0.9: soft_EM=1
else: soft_EM = 0

- **F1-Score**: The average token overlap between predicted (system answers) and gold answers (ground truth).
  - $\text{Precision} = \frac{\text{Number of words in the predicted and gold answers}}{\text{Total number of words in the predicted answer}}$
  - $\text{Recall} = \frac{\text{Number of words in the predicted and gold answers}}{\text{Total number of words in the gold answer}}$.

- **Mean Reciprocal Rank (MRR)**: The avg. rank at which the first relevant answer appears across a set of queries.

$$\text{MRR} = \frac{1}{|Q|}\sum_{i=1}^{|Q|}\frac{1}{rank_i}$$

Example of QA Metrics

What color is the cat in this picture?

Options:

A: White cat

B: Brown cat

C: The black cat [CORRECT]

D: Yellow cat

Probabilistic predictions: A=0.45, B=0.20, C=0.10, D=0.25

Sorted predictions: A=0.45, D=0.25, B=0.20, C=0.10

Predicted Answer: A=White cat (0.45)

Exact Match (EM)=0

Reciprocal Rank (RR)=0.25

Precision=1/2=0.5, Recall=1/3=0.333

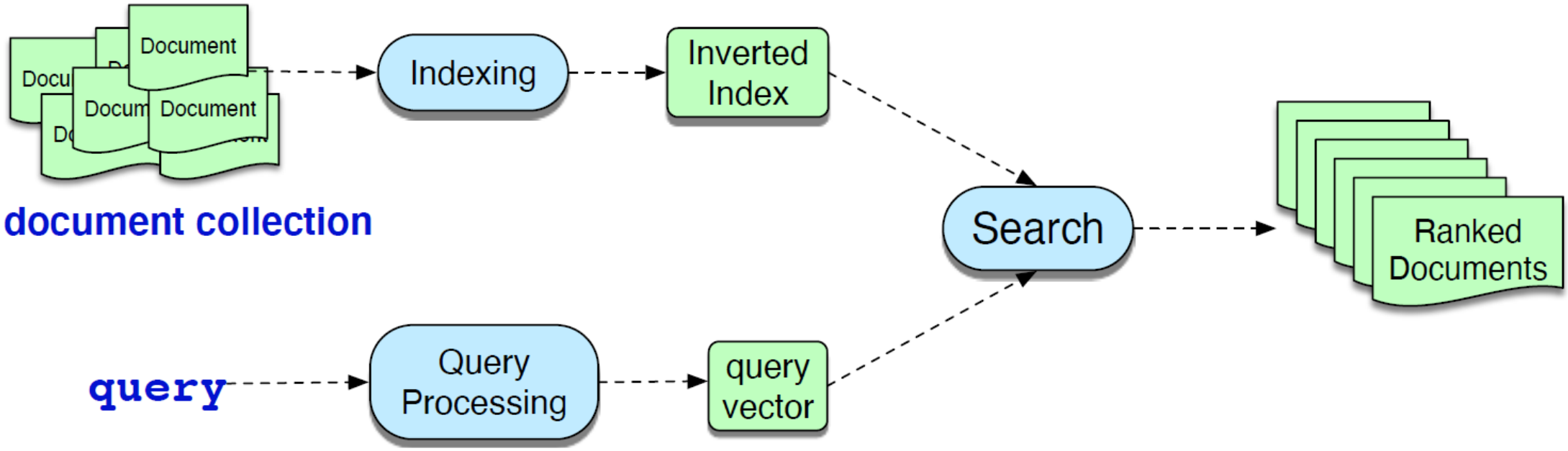
N.B. This is a hypotetical example of a visual question – there it no explicit picture required above.

Today

- Question Answering
- **Information Retrieval**
- Retrieval Augmented Generation

Acknowledgement: Some images from https://web.stanford.edu/~jurafsky/slp3/11.pdf

Traditional Information Retrieval



- Information Retrieval (IR) is the field of study concerning the ranking and retrieval of documents to a user query.
- The documents can be of different types such as web pages, scientific papers, news articles, audio/video/image files, etc.

Term Frequency for Document Scoring

- Term frequency (**tf**):

$$\mathrm{tf}_{t,d} = \begin{cases} 1 + \log_{10} \mathrm{count}(t,d) & \text{if } \mathrm{count}(t,d) > 0 \\ 0 & \text{otherwise} \end{cases}$$

- Inverse Document Frequency (**idf**):

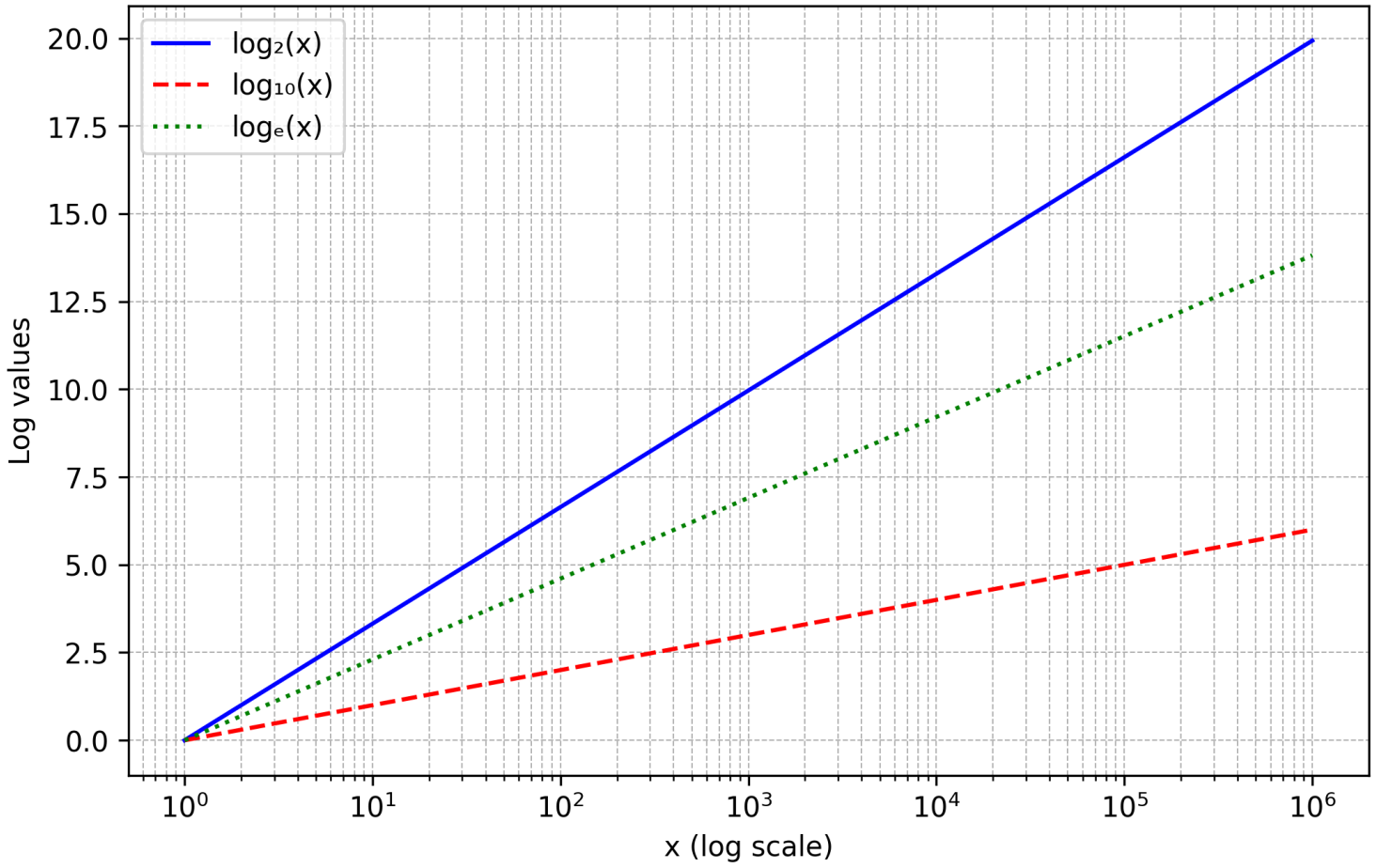
$$\mathrm{idf}_t = \log_{10} \frac{N}{\mathrm{df}_t}$$

- Term frequency - Inverse Document Frequency (**tf-idf**):

$$\text{tf-idf}(t,d) = \mathrm{tf}_{t,d} \cdot \mathrm{idf}_t$$

Symbols: t=word, d=document, N=total number of docs., df_t number of docs. *t* occurs in, $\mathrm{tf} - \mathrm{idf}(t, d) \geq 0$.

Logarithmic Values



Since raw term frequency (x-axis) can result in large values, algorithmic values (y-axis) are used instead.

Document Scoring via Cosine Similarity

- We score document *d* and query *q* by the cosine of their vectors **d** and **q**:

$$\text{score}(q,d) = \cos(\mathbf{q},\mathbf{d}) = \frac{\mathbf{q}\cdot\mathbf{d}}{|\mathbf{q}||\mathbf{d}|}$$

- Using tf-idf values we get

$$\text{score}(q,d) = \sum_{t\in\mathbf{q}} \frac{\text{tf-idf}(t,q)}{\sqrt{\sum_{q_i\in q}\text{tf-idf}^2(q_i,q)}} \cdot \frac{\text{tf-idf}(t,d)}{\sqrt{\sum_{d_i\in d}\text{tf-idf}^2(d_i,d)}}$$

Notation: bold letters denote vectors and |.| mean 'vector length', magniture or norm, e.g., $|q| = \sqrt{\sum_{i=1}^{N} q_i^2}$

Example of Document Scoring Using tf-idf and Cosine Similarity

Query:	sweet love
Doc 1:	Sweet sweet nurse! Love?
Doc 2:	Sweet sorrow
Doc 3:	How sweet is love?
Doc 4:	Nurse!

1. Define the vocabulary (unique words in the query and docs).
2. Calculate the following scores for the query and each of the documents: tf, idf, df, tf-idf.
3. Apply cosine similarity to the query and each of the documents.

Example of Document Scoring Using tf-idf and Cosine Similarity

Query						
word	**cnt**	**tf**	**df**	**idf**	**tf-idf**	**n'lized** = tf-idf/$\|q\|$
sweet	1	1	3	0.125	0.125	0.383
nurse	0	0	2	0.301	0	0
love	1	1	2	0.301	0.301	0.924
how	0	0	1	0.602	0	0
sorrow	0	0	1	0.602	0	0
is	0	0	1	0.602	0	0
$\|q\| = \sqrt{.125^2 + .301^2} = .326$						

	Document 1					Document 2				
word	**cnt**	**tf**	**tf-idf**	**n'lized**	**× q**	**cnt**	**tf**	**tf-idf**	**n'lized**	**×q**
sweet	2	1.301	0.163	0.357	**0.137**	1	1.000	0.125	0.203	**0.0779**
nurse	1	1.000	0.301	0.661	0	0	0	0	0	0
love	1	1.000	0.301	0.661	**0.610**	0	0	0	0	**0**
how	0	0	0	0	0	0	0	0	0	0
sorrow	0	0	0	0	0	1	1.000	0.602	0.979	0
is	0	0	0	0	0	0	0	0	0	0
$\|d_1\| = \sqrt{.163^2 + .301^2 + .301^2} = .456$						$\|d_2\| = \sqrt{.125^2 + .602^2} = .615$				
Cosine: $\sum$ of column:					**0.747**	Cosine: $\sum$ of column:				**0.0779**

Alternative to tf-idf: BM25

The BM25 score between query q and document d is

$$\sum_{t \in q} \overbrace{\log\left(\frac{N}{df_t}\right)}^{\text{IDF}} \overbrace{\frac{tf_{t,d}}{k\left(1-b+b\left(\frac{|d|}{|d_{\text{avg}}|}\right)\right)+tf_{t,d}}}^{\text{weighted tf}}$$

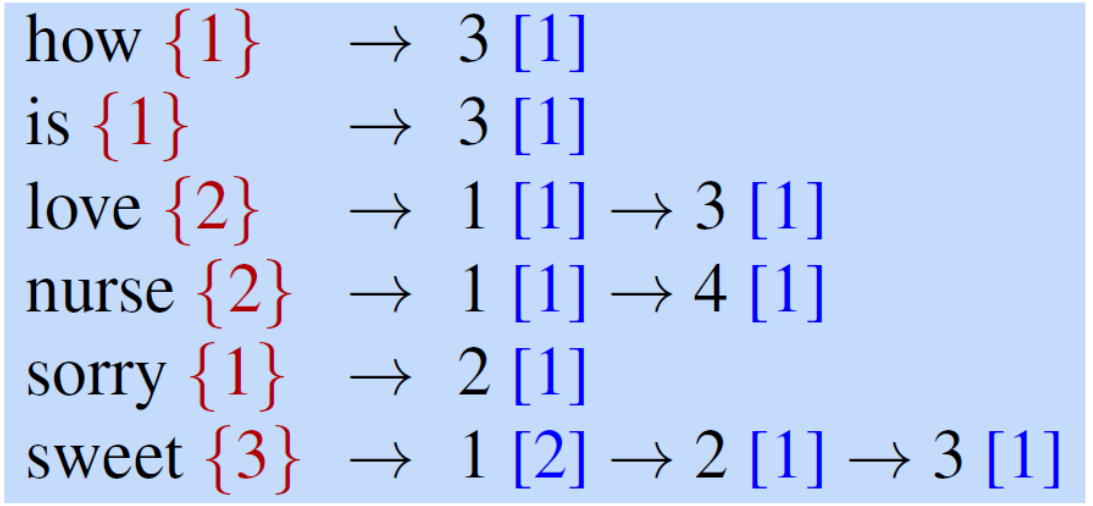
where $|d_{avg}|$ is the average the length of documents in the data, b is a scaler for the document length, and k controls how much term frequency (tf) contributes to the score. Typical values of k and b: $k = [1.2, \ldots, 2]$ and $b = 0.75$.

Source of values above: Manning, et al. ``Introduction to Information Retrieval''. Cambridge U. Press, 2008.

Inverted Index

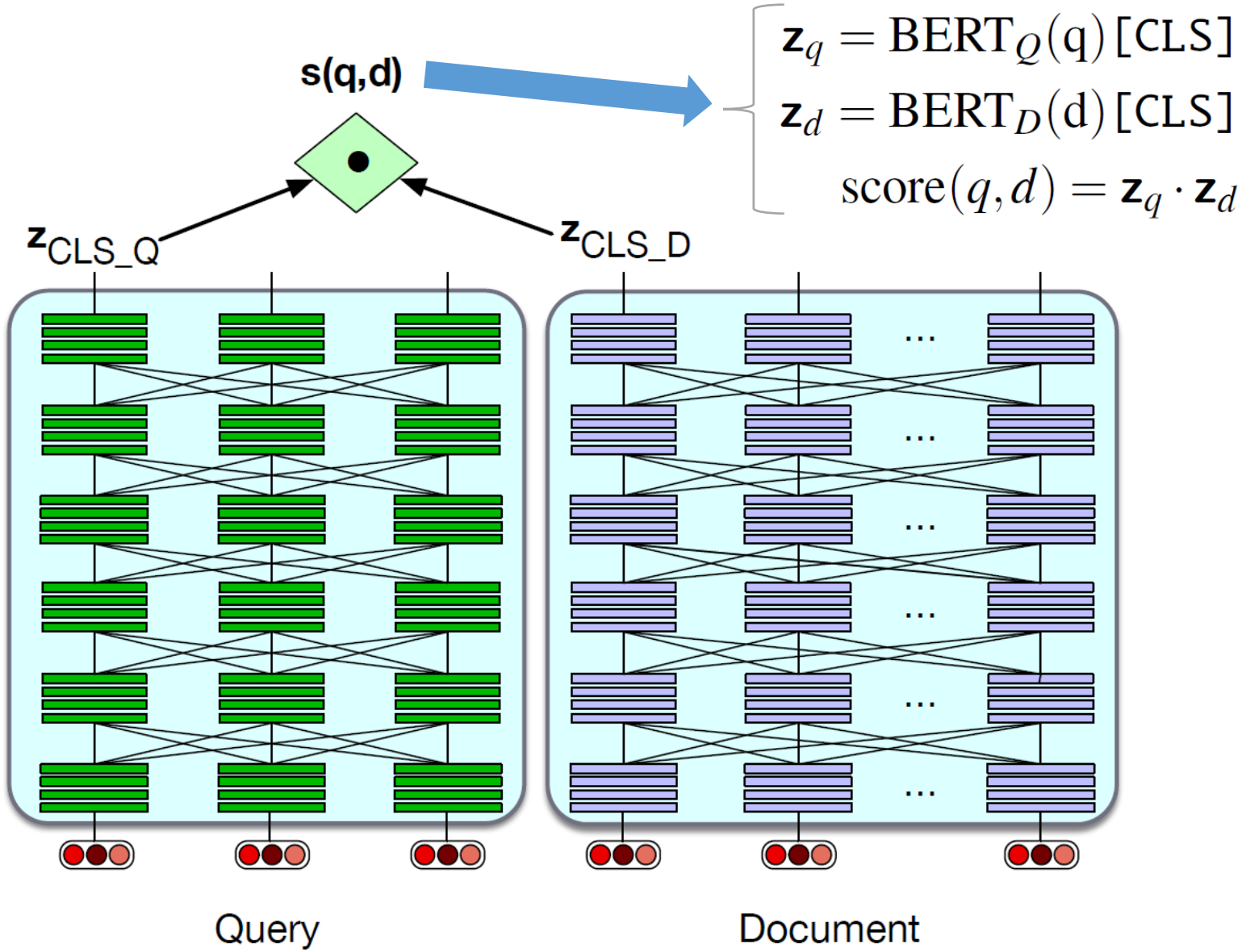
This is a data structure used to find documents efficiently, which contains a dictionary and the postings.

N-grams can be used instead of words/tokens.



The dictionary is a list of terms (words) and a posting list is the list of document IDs associated to each vocabulary item containing information such as the term frequency.

Information Retrieval (IR) with Dense Vectors



Evaluation of IR Systems

$$Precision = \frac{|R|}{|T|} \qquad Recall = \frac{|R|}{|U|}$$

- $|R|$: the number of relevant documents retrieved (true positives).
- $|T|$: is the number of retrieved/ranked documents (including relevant and irrelevant = true positives + false positives).
- $|U|$: is the number of all relevant documents (retrieved and not retrieved = true positives + false negatives).

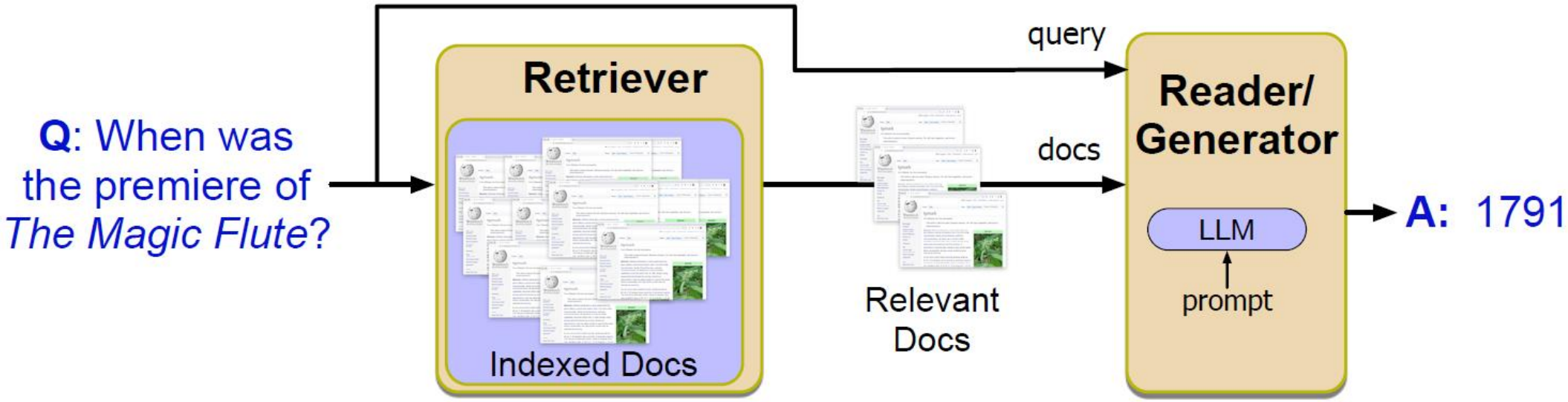
See slides of week 3 for a combination of precision and recall, metric referred to as F1-Score.

Today

- Question Answering
- Information Retrieval
- **Retrieval Augmented Generation**

Acknowledgement: Some images from https://web.stanford.edu/~jurafsky/slp3/11.pdf

Question Answering (QA) via Retrieval Augmented Generation



Retriever: returns relevant documents from a collection.

Generator: provides a response given a prompt and relevant docs.

Note that using RAG corresponds to an OPEN book QA task and not using RAG to a CLOSED book QA task.

Question Answering via Retrieval Augmented Generation (RAG)

Schematic of a RAG Prompt

```
retrieved passage 1

retrieved passage 2

...

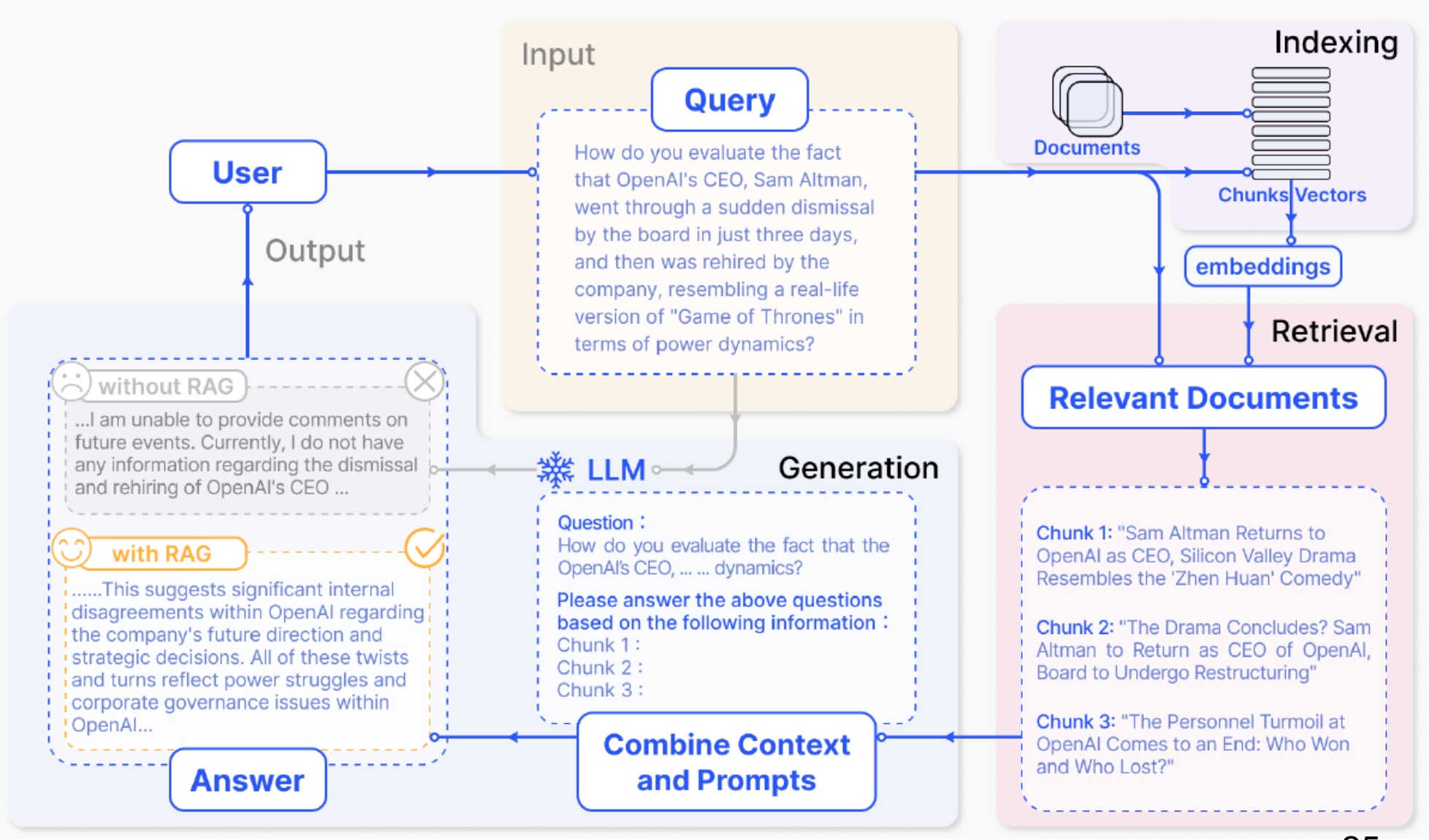
retrieved passage n

Based on these texts, answer this question:  Q: Who wrote
the book ‘‘The Origin of Species"?  A:
```

More formaly

$$p(x_1, \ldots, x_n) = \prod_{i=1}^{n} p(x_i | \mathrm{R}(q)\,;\,\text{prompt}\,;\,\texttt{[Q:]}\,;\,q\,;\,\texttt{[A:]}\,;x_{<i})$$

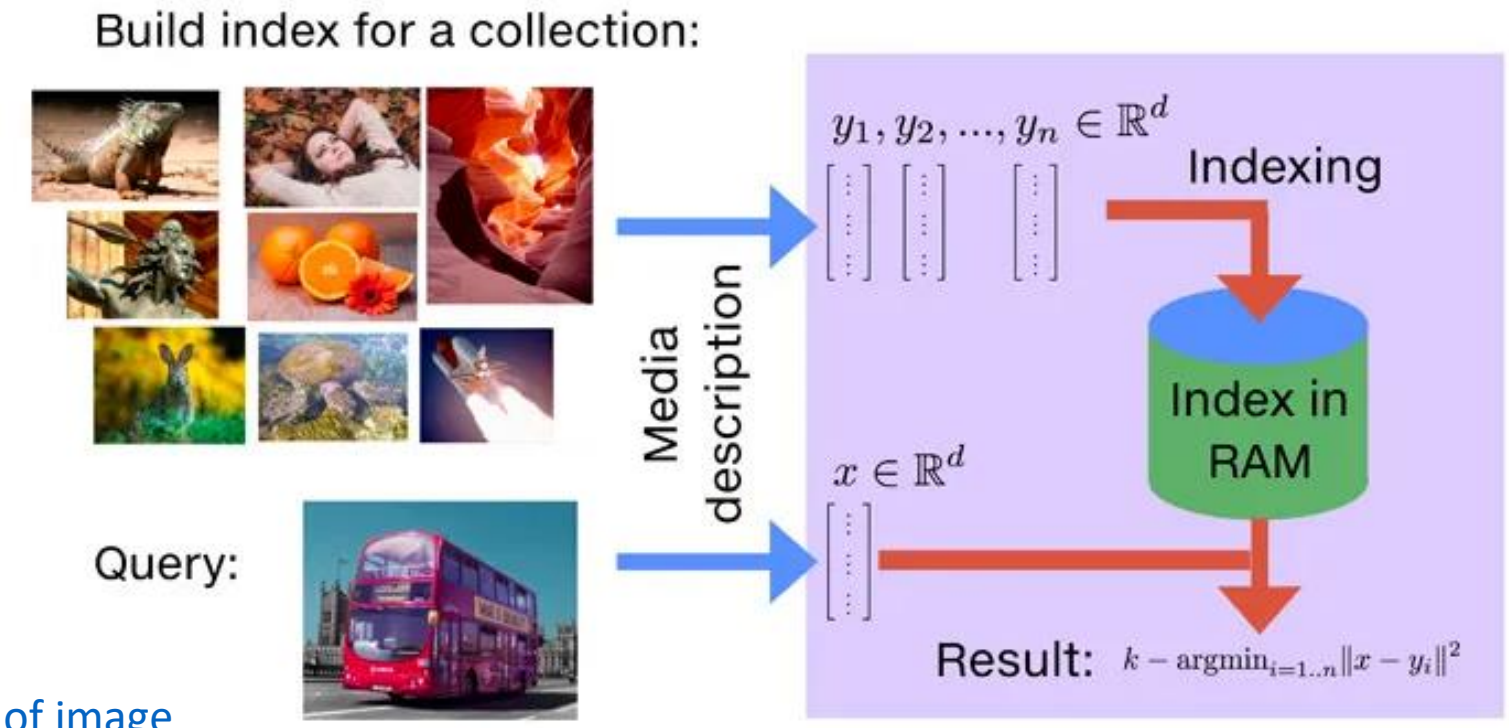
RAG with Frozen LLMs



Gao, et al. ``Retrieval-Augmented Generation for Large Language Models: A Survey''. CoRR, 2024.

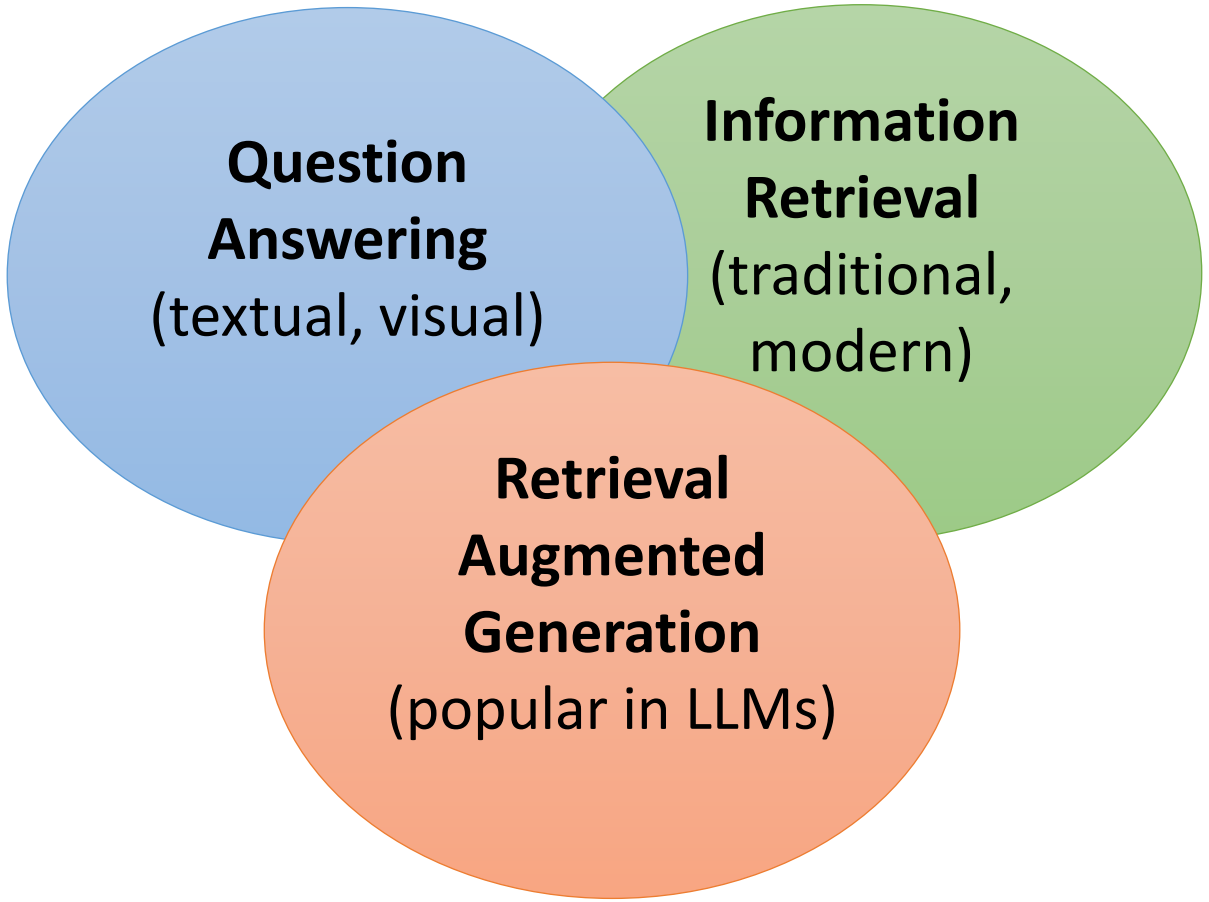
Efficient Similarity Search for RAG

The library Facebook AI Similarity Search (FAISS) returns a list of database objects that are nearest to a query vector in terms of Euclidean distance—optimised for large scale searches.



source of image

Today


Question Answering
(textual, visual)
Information Retrieval
(traditional, modern)
Retrieval Augmented Generation
(popular in LLMs)

What's Next?

- Readings:
  - Jurafsky and Martin. Speech and Language Processing. Chapter 11, 2026. [recommended]
  - Zhu, et al. "Large Language Models for Information Retrieval: ASurvey". ACM Trans. Inf. Sys., 2026.

- Workshop **tomorrow**: Prompting LLMs with RAG, Visual Question Answering, and Metrics.

- Next lecture: Constituent and Dependency Parsing